

Red protein suppresses breast-cancer lung metastasis

It docks on the outer mitochondrial membrane, holds DRP1 away from the green partner, and thereby shuts off fission-induced UPRmt, nuclear EMT signaling, and lung colonization.

1 Lung metastasis is suppressed
in breast cancer

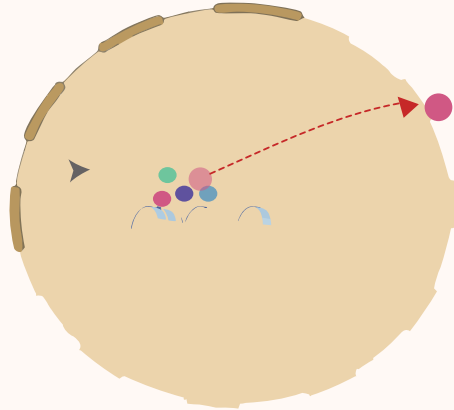
2 Outer membrane: DRP1 is sequestered
no green-partner binding, no fission, no UPRmt

3 EMT genes stay off
mitochondrial proteins do not enter the nucleus

A On the mitochondrion

Mitochondrial pool

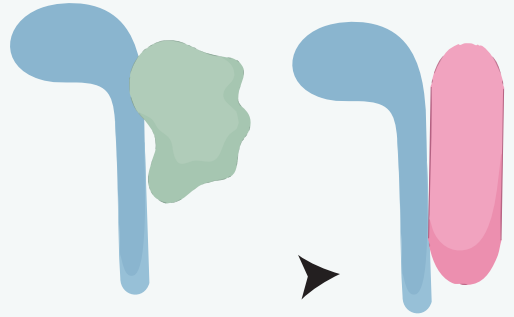
Outer-membrane form



The red protein resides in mitochondria and can sequester DRP1 when fission is blocked

B At the outer membrane

OUTER MITOCHONDRIAL MEMBRANE



Red protein + DRP1 + Green protein

Red protein binds DRP1 and blocks its contact with the green partner, so DRP1 cannot oligomerize.

suppresses the cascade below

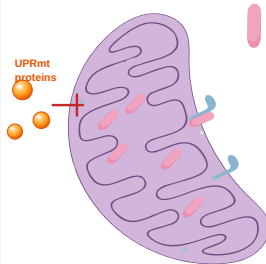
C Fission and UPRmt



BLOCKED

DRP1 would assemble on the green protein and split UPRmt stays off.

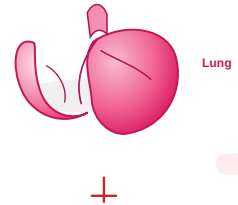
D Nucleus and EMT



BLOCKED

UPRmt would carry mitochondrial proteins into the nucleus and EMT stops.

E Lung metastasis



BLOCKED

EMT makes these cells migratory. They colonize the lung.

LEGEND · drawings are from the original figure

Red protein



DRP1
dynamin-related protein 1



Green protein
DRP1-binding partner



UPRmt-associated protein
moves to the nucleus if UPRmt is on



Inhibition

Red protein is the brake. Blocking DRP1 stops fission, UPRmt, EMT transcription, and lung metastasis.